

(WIP) Homeownership: First Time Buyer Signals

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💡 **WIP:** This document covers ongoing work to identify and target First Time Buyers (FTBs) roughly 6 months before purchase. The goal is to surface relevant, timely app content and successfully guide users into the FTB funnel.

❓ What's the problem?
📋 What can we do instead?
- 🍱 Bento
- 🏘️ Home Screen Optimisation

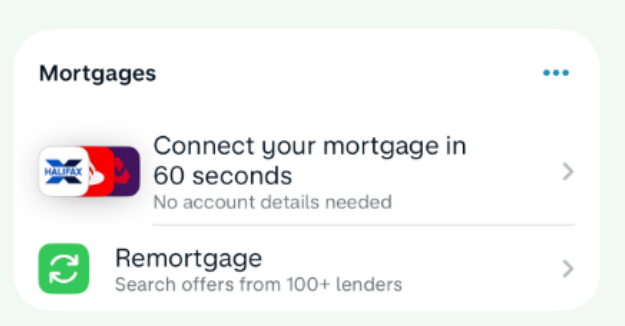
📊 What data can we use? (our "Target" variable)
- 📈 Primary signal: batch data going from 0 to 1

👥 Initial Features
👩‍🔬 Initial Results
- **Key metrics**

📢 Rule-based Strategies
⏭️ Next Steps

❓ What's the problem?

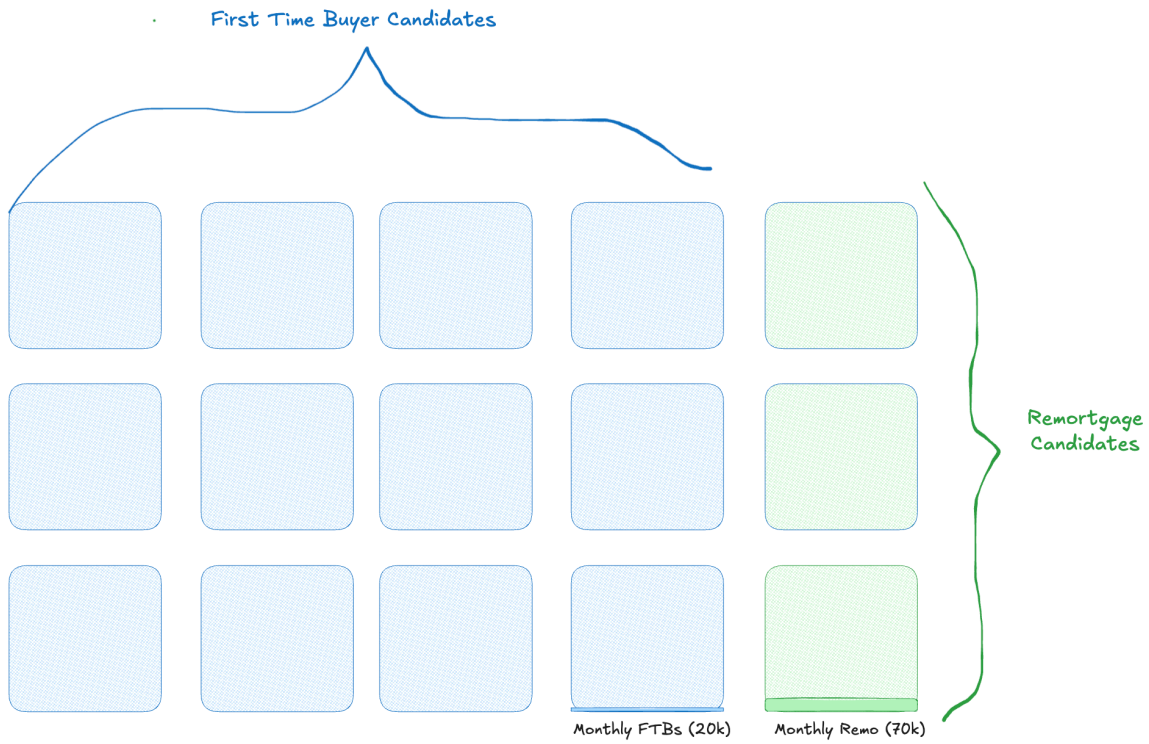
Today, Homeownership gets ~60% of remortgage referrals from the Monzo Home Screen entry point.**(Source)**



This entry point is shown to all users with a mortgage (3 million), of which 2.4% (~1/42) are actually in-market. *This entry point is irrelevant to ~97% of users that it reaches.*

The issue is that the Home Screen is effectively “always on”: it reaches ~3m mortgage-holders, but only ~2.4% are in-market at any point in time (i.e. low “precision”). So the surface is irrelevant for ~97% of the people who see it.

As we plan to open up to First Time Buyers, this becomes a bigger problem: we don’t currently have a scalable way to identify and target FTBs in-app. If we went for a similar approach (an overview entry point for all “eligible” customers), we’d be targeting 12million customers, of which we estimate ~25k become a first time buyer each month. This would be an irrelevant surface to 99.8% of customers. It’s unlikely that we’d be allowed to use the overview screen in this way, and it’s debatable whether we’d even want to.



70k/3million Remortgage candidates come into market per month (~2.3%). For First Time Buyers, we have ~25,000/12 million, which is ~0.17%.

TLDR: We need to build a scalable, durable way of surfacing FTB entry points in as relevant a way as we can.

📝 What can we do instead?

🍱 Bento

MonzOS/The App Personalisation Squad are working on **Bento,** a recommendation system/grid UI component to replace **Suggested For You**, and the **Do More menu.** This will eventually be surfaced across Monzo, and aims to show the right content to the right users at the right time.

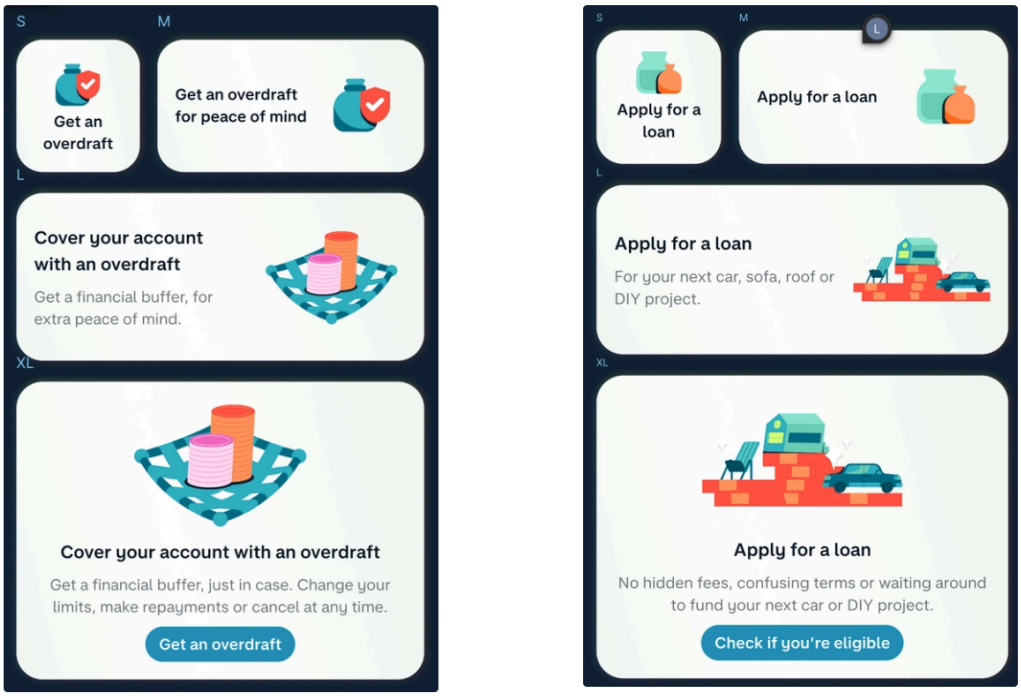
It’s a model built off of a big list of user features, which is able to learn in real time the type of users that are interested in given content. It’s currently prioritises content which users are most likely to “convert” through, and eventually will take into account the associated LTV of a given action.

💡 **Prioritisation Explainer**

- Currently, if the model believes that a user is 5 x more likely to take out Flex than they are to remortgage, this will get prioritised
- In future, remortgage could be prioritised over Flex instead, provided the expected revenue of a remortgage compensates for the lower conversion likelihood.

We can work with the MonzOS team to get new features added to this model, to enable it to start learning which users to show FTB/Remo content to.

Bento is aiming for release at the end of June.



An example of what the Bento UI looks like.

A WIP notion doc I found about Bento

🏘️ **Home Screen Optimisation**

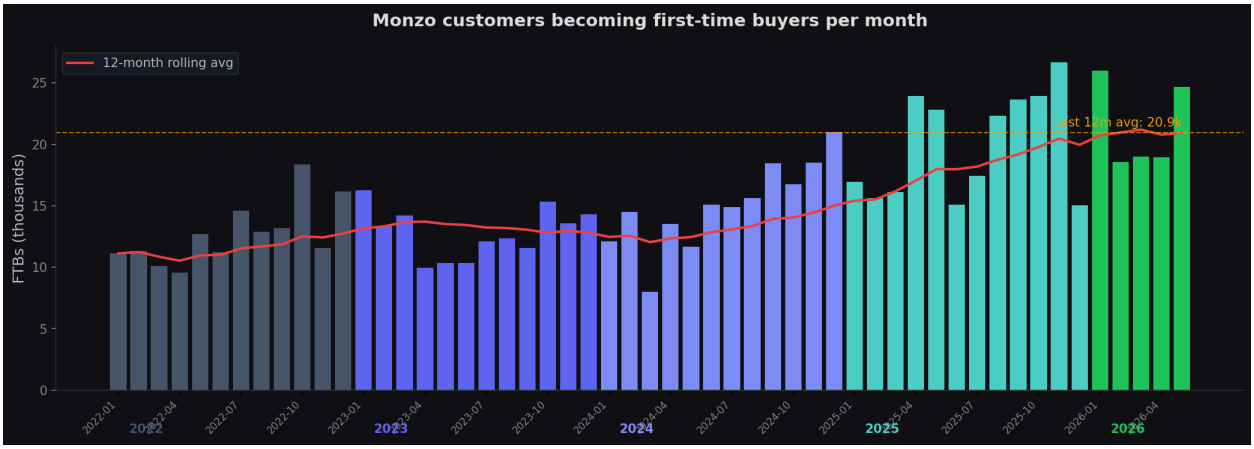
We currently have about 2.4% precision on our remortgage home screen targeting. If we can define a set of targeting rules that can achieve something similar for First Time Buyers, we might be able to justify doing something similar.

N.B. in general it's expected that the home screen will move away from the current layout of multiple teams using it for "always on" entry points - so this shouldn't be seen as a long-term solution.

📊 What data can we use? (our “Target” variable)

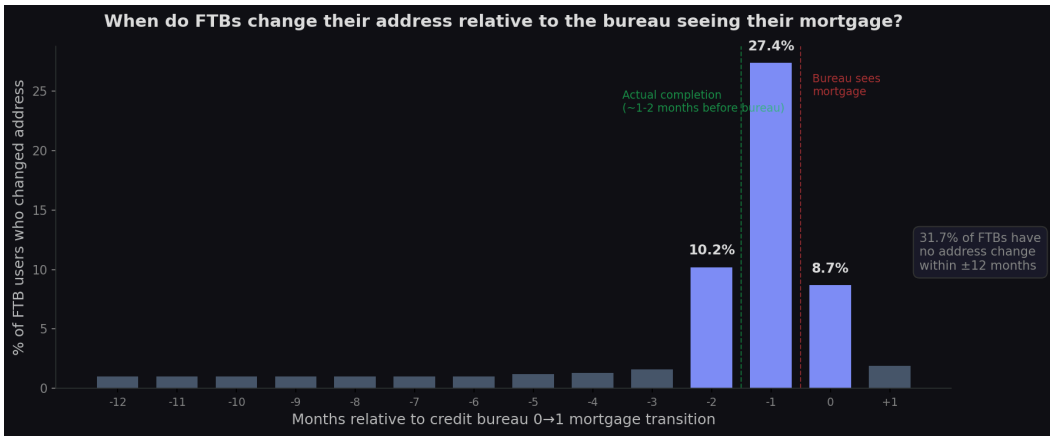
📈 Primary signal: batch data going from 0 to 1

We can see which existing customers have gone from having 0 mortgages to 1 mortgage, and when.



We can see our current 12 month average of First Time Buyers is ~21k per month.

We can compare this to address change data to get rough signal for when the actual mortgage started:



We see a big spike in address changes 1 month before the batch data updates. We also see a lot of address changes at 2 months before.

This modelling currently assumes a 2 month lag. For a given customer in a given month, we ask the question:

Did this customer buy their first house between 6 and 10 months from now?

Training data structure: one row per user per month

user_id	observation_month	becomes_ftb_in_8_to_12m
user_000...	2022-01	Yes
user_000...	2025-06	Yes
user_000...	2026-01	No
user_000...	2026-05	No
user_000...	2025-10	No

In practise, we have millions of customers, with years worth of data. For efficiency, we are initially looking at a sample of 300,000 users, split equally into the following buckets:

1️⃣ **Bucket 1: Users that buy a house 6-10 months from now**

2️⃣ **Bucket 2: Users that buy a house eventually, but not 6-10 months from now**

3️⃣ **Bucket 3: Users that have never bought a house in batch data (*they still might one day)***

This allows the model to learn to discriminate between users with FTB type features, as well as discriminate between the timing of users that are FTB soon vs eventually.

(The initial model only looked at FTBs 6-10 months from now vs people that never FTB, so was pretty good at predicting them, but didn't do well when it eventually saw users from Bucket 2)

ℹ️ **Current Limitation:**

We'll look to iterate on the model based on connected customers deal start dates (where it's their first mortgage). This will take away the guesswork associated with the batch data's lag.

👥 Initial Features

So far, 61 features are measured at the observation month, grouped by category. All features are point-in-time, to prevent leakage.

Demographics

Feature	Type	Hypothesis
Age	Continuous	FTBs cluster in late 20s to mid 30s — the typical first-time buying age.
Is London	Binary	London is more expensive, so FTBs there may show different patterns or be less likely at any given time.
Lives with others	Binary	Address connections > 1. Proxy for cohabiting with a partner — couples may be more likely to buy together.

Intent signals

Feature	Type	Hypothesis
Has house-named pot	Binary (PII)	Pot named "house", "deposit", "mortgage", "first home", "stamp duty", etc. Explicit saving intent.

Feature	Type	Hypothesis
House pot age (months)	Continuous	How long they've been saving with intent — longer pot age could mean closer to purchase or longer planning horizon.
Has ISA pot	Binary	Tax-advantaged saving. LISA specifically is for FTBs, but we can't isolate it from generic ISA.

Financial readiness

Feature	Type	Hypothesis
Credit score	Continuous	Need a decent score to get a mortgage. Higher score = more mortgage-ready.
Days since credit search	Continuous	Recent hard credit search may indicate mortgage application activity.
Has verified income	Binary	Went through income verification — financially engaged with Monzo.
Verified income (£)	Continuous	Higher income = greater mortgage affordability.
Declared annual income (£)	Continuous	Self-declared when applying for Monzo credit products. Could indicate borrowing behaviour rather than buying.
Debt to income	Continuous	Existing debt reduces mortgage capacity — higher ratio may mean less likely to buy soon.
Has joint account	Binary	Couples opening a joint account may be consolidating finances ahead of a property purchase.

Savings

Feature	Type	Hypothesis
Total pot balance (£)	Continuous	Larger savings = closer to having a deposit.
Max single pot balance (£)	Continuous	One large pot likely represents dedicated deposit savings.
Pots over £5k / £10k / £25k	Count	Threshold-based deposit readiness signal — more large pots = more liquid savings.
Total open pots	Count	Active savers use more pots — proxy for financial organisation.
ISA balance (£)	Continuous	Tax-advantaged savings. Larger ISA balance could indicate long-term deposit saving.

Savings velocity

Features computed at multiple lookback periods to capture savings trajectory.

Feature	Type	Hypothesis
Savings velocity 1m / 3m / 6m / 12m	Continuous	Pot balance growth over period. Accelerating savings could indicate active deposit building.
Savings acceleration 3m / 6m	Continuous	Is growth speeding up? Could indicate intensifying saving effort as purchase approaches.
Pot count change 3m / 6m	Continuous	Opening new pots may signal new savings goals (e.g. stamp duty, solicitor fees).

Income dynamics

Feature	Type	Hypothesis
Income change 3m / 6m	Continuous	A pay rise increases mortgage affordability and may trigger the decision to buy.

Spending

Feature	Type	Hypothesis
Card spend (30d)	Continuous	Baseline spending level — context for other features.
Outbound amount (30d)	Continuous	Total outbound transfers including rent, bills, etc.
Card spend change 3m / 6m	Continuous	Cutting back on spending could indicate saving more aggressively for a deposit.
Outbound change 3m / 6m	Continuous	Changes in outbound transfers — could capture rent stopping or new large payments.

Direct debits

Feature	Type	Hypothesis
Active DDs	Count	More DDs = more established financial life (bills, subscriptions, insurance).
DD amount (30d)	Continuous	Total DD outflow — proxy for fixed monthly commitments.
DD count change 1m / 3m / 6m	Continuous	Cancelling DDs may signal a move (cancelling utilities, gym, council tax at old address).
DD amount change 1m / 3m / 6m	Continuous	£ change in DD commitments — new DDs for a new property or cancellations from an old one.

Standing orders

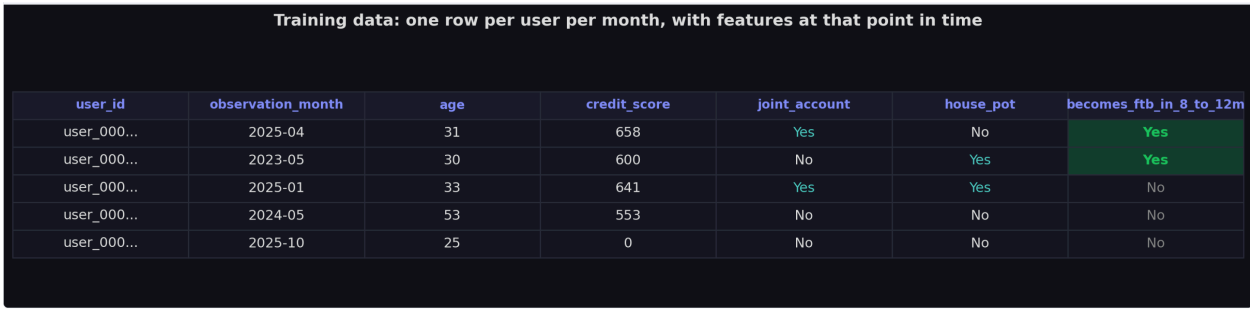
Feature	Type	Hypothesis
Active SOs	Count	Standing orders at observation date — rent payments often set up as SOs.
Active SO amount (£)	Continuous	Total SO outflow. Large SO could be rent — relevant context for affordability.
SOs cancelled in 3m / 6m (count + £)	Continuous	Cancelling a standing order could indicate stopping rent payments ahead of a move.

Investments & long-term savings

Feature	Type	Hypothesis
Investment balance	Continuous	Users with investments may liquidate to fund a deposit.
Investment net inflows	Continuous	Are they still investing or withdrawing? Net outflow could signal deposit gathering.
Has investment auto-deposit	Binary	Automated investment saving — financially engaged users.
Investment balance change 3m / 6m	Continuous	Drawdown of investments ahead of purchase.
Fixed savings balance	Continuous	Fixed-term savings — may need to wait for maturity before buying.
Fixed savings change 3m / 6m	Continuous	Withdrawing fixed savings to fund deposit.
ISA net inflows	Continuous	Continuing to save in ISA — long-term deposit building.
ISA withdrawals amount	Continuous	Withdrawing ISA funds — could indicate deposit gathering.

Credit engagement

Feature	Type	Hypothesis
Credit insights engaged	Binary	Actively checking credit score in-app — may indicate checking readiness for a mortgage



👩🏻‍🔬 Initial Results

We fed these features into a machine learning model (XGBoost) which learns the patterns that distinguish future FTBs from non-buyers. It combines all 61 features together - weighting and interacting them - to produce a single score for each user: *how likely they are to become a first-time buyer (in batch data) the next 8-12 months.*

All results below are from a **representative sample of ~490k random non-mortgage users**.

Key metrics

Metric	Value	Meaning
AUC	**0.82**	If you take a random FTB and a random non-FTB, the model scores the FTB higher 82% of the time
Top decile lift	**4.0x**	We can quadruple our “relevance/precision” if we look at the top 10% of FTBs from the models output (from 0.17% to 0.67%
FTBs captured in top 10%	**40%** of all FTBs	We’d be reaching 10% of users (1.2mil instead of 12mil), but would still reach 40% of FTB (8k/20k)
Monthly base rate	**0.17%** (20k FTBs / 12M users)	This is our “base level” relevance if we did no targeting and went to everyone

The model ranks most FTBs above most non-FTBs (AUC 0.82) and concentrates them at the top of the score distribution.

How the model concentrates FTBs

Decile	Users	FTB rate	Lift vs random
Top 10%	**1,200,000**	**0.67%**	**4.0x**
2nd 10%	1,200,000	0.43%	2.6x
3rd 10%	1,200,000	0.26%	1.5x
4th 10%	1,200,000	0.13%	0.8x

Decile	Users	FTB rate	Lift vs random
5th 10%	1,200,000	0.07%	0.4x
Bottom 50%	6,000,000	0.03%	0.1x

What this means in practice

Target group	Users shown content	FTBs reached/month	Not relevant for	Relevance rate
No model (everyone)	12,000,000	20,000	11,980,000	0.17%
Top 20%	2,400,000	13,200	2,386,800	0.55%
Top 10%	**1,200,000**	**8,000**	**1,192,000**	**0.67%**
Top 5%	600,000	4,000	596,000	0.67%
Top 1%	120,000	1,100	118,900	0.95%

For context, the current remortgage approach targets 3M users at **2.3% relevance rate**. The FTB model doesn't reach that level because the FTB base rate (0.17%) is structurally much lower. However, the model provides **4-6x lift** over random, and top-scored users are disproportionately engaged Monzo power users - **so actual conversion on referrals is likely higher than raw relevance rate suggests** 🔑32

What drives predictions

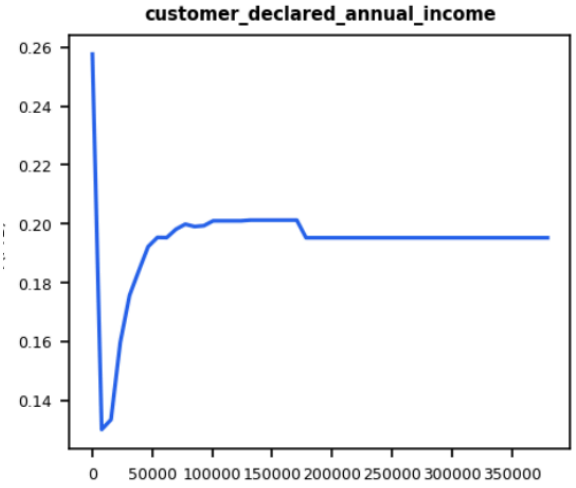
Top features

Feature	Importance	What it means
Declared annual income	0.079	**Inverse signal.** Having declared income means the user applied for Monzo credit (Flex/loans) — borrowing mode, not buying mode.
Days since credit search	0.077	Recent hard credit search suggests mortgage application activity.
Credit score	0.072	Higher credit score = more mortgage-ready.
Joint account	0.061	Couples consolidating finances — classic pre-purchase signal.
Income change (6m)	0.052	A pay rise increases affordability and may trigger the decision to buy.
Age	0.045	FTBs cluster 28-35.
Verified income	0.040	Has gone through Monzo income verification — financially engaged.
Is London	0.033	London residents less likely to FTB at any given time (harder to buy).
House-named pot	0.031	Explicit intent — pot named "house", "deposit", "mortgage", etc.

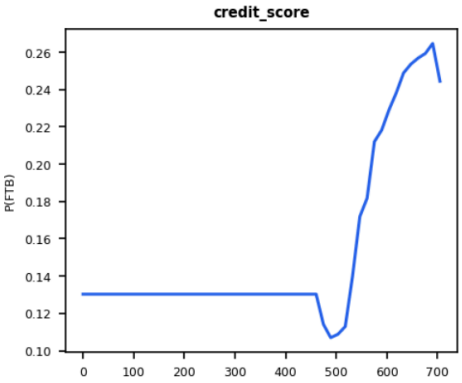
No single feature dominates. The model uses a balanced mix of demographic, financial, intent, and credit signals.

📈 Partial Dependency Plots

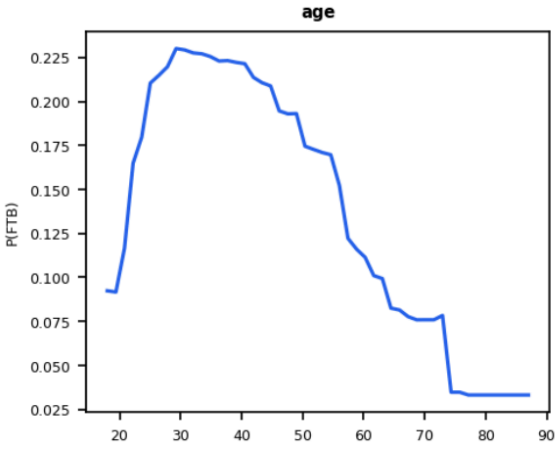
For a given feature, we can plot how the models predictions will change if that given features changes. ***N.B.*** *this does not take interactions between features into account.*



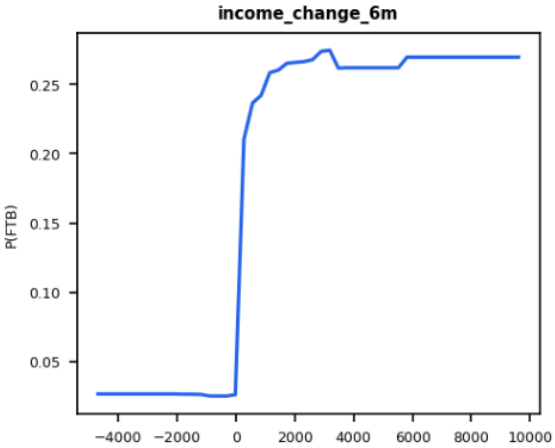
The declared income here implies that a customer has applied for a borrowing product. 0 here means they have not applied fro a product, which will correlate with wanting a mortgage soon.



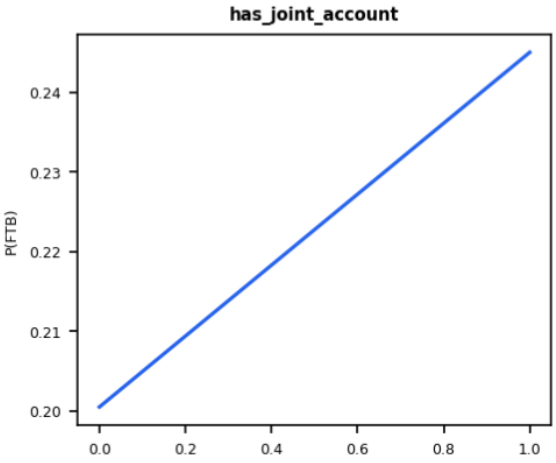
The model won't have seen credit scores below a certain number, and will have seen 0s when a customer doesn't have a credit score in the bureau data. We then see a small dip for low credit scores, before a steep increase in propensity based on score.



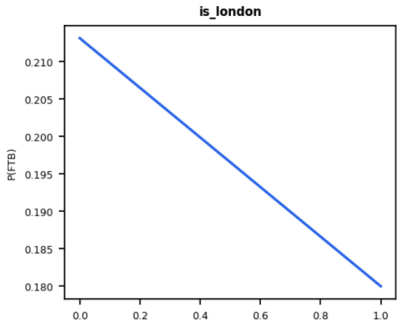
We see a ramp up from 20 to 30, and then a slow decline



A recent payrise is a strong signal of a future mortgage.



Users with joint accounts are more likely to first time buy



Users in London are less likely to first time buy

📢 Rule-based Strategies

We can also look at using learnings from the model to build out some simple rules to apply to target customers. (I.e., instead of just `Has no mortgage`, we can do `Has no mortgage` + `has 10k saved` + `has a joint account`)

We can look at how different rules perform on the representative population, scaled to 12M non-mortgage users and 20k FTBs/month.

Baseline: showing to all 12M users reaches all 20k FTBs/month at 0.17% relevance rate

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Rules vs ML model

Approach	Users Reached	FTBs/month	Relevance	Compared to baseline	Recall	Engagement score
No targeting	12,000,000	20,000	0.17%	**1 x**	100%	—
`Age 25-35` + `credit > 550`	3,200,000	11,400	0.36%	**2.1 x**	57%	0
`Joint account` only	650,000	3,400	0.52%	**3 x**	17%	1
`Age 25-35` + `joint account`	387,000	2,500	0.65%	**3.8 x**	13%	1
`Age 25-35` + `credit > 550` + `joint account`	305,000	2,200	0.72%	**4.2 x**	11%	1
`Joint account` + `house pot`	134,000	960	0.73%	**4.3 x**	5%	2
`Joint account` + `credit > 550` + `no declared income`	132,000	1,100	0.82%	**4.8 x**	5%	1

Approach	Users Reached	FTBs/month	Relevance	Compared to baseline	Recall	Engagement score
`Joint account` + `house pot` + `age 25-35`	87,000	700	0.80%	**4.7 x**	4%	2
`House pot` + `balance > £5k`	52,000	460	0.87%	**5.1 x**	2%	2
ML model top 10%	**1,200,000**	**8,000**	**0.67%**	**4 x**	**40%**	
ML model top 5%	**600,000**	**4,000**	**0.67%**	**4 x**	**20%**	
ML model top 1%	**120,000**	**1,100**	**0.95%**	**5.6 x**	**6%**	

👀 **What's the engagement score?**

It's one thing for the entry point to be "relevant" to a customer (i.e., they are genuinely looking for a house), but it's another thing for the customer to be "the type of customer that will be interested in going through the process with Monzo".

We can hypothesise that "Monzo Power Users" are more likely to actually click on a relevant entry point if they see one. **A next step here is to build a similar model for "referral propensity", which we can use our remortgage referral data for.** I.e. can we be even more targeted with customers who we think want to buy a house, and want to do it with Monzo?

For now, we've just made a simple score based on the number of "engagement signals" we have in a rule:

Feature	Score	Reasoning
Joint account	1	Opted into a Monzo product
House-named pot	1	Actively using pots with intent
Pot balance > £5k	1	Meaningful savings on Monzo
Age	0	Demographic
Credit score	0	External (bureau)
Declared income	0	External (self-reported)

The engagement score is the sum of product-adoption features in the rule. Higher score = the rule selects for users who actively use Monzo, which we hypothesise correlates with higher click-through on mortgage content. This can be validated with a separate referral propensity model.

⏭️ Next Steps

- Define + prioritise **new signals/features** we could build to improve the models precision
- Model tuning
- Look into a similar piece of analysis on referral propensity - *can we target FTBs with high intent/Monzo affinity?*
- Partner with **MonzOS** to get the chosen signals **added to RecTrack** (so they're available for targeting)
- Design and run an experiment comparing a **targeted entry point** (using FTB propensity / rules based targeting) vs. a **generic entry point**